Public Sentiment Analysis Toward Official COVID-19 Publications Using Natural Language Processing (NLP) Techniques

IE7500 Group10: Ruthie Orona, Geovany Silva, Luce Trinh, Sabrina Valcin

1. Project Description

The objective of this project is to analyze the reactions and comments expressed by users on social media in response to official publications issued by public health organizations, primarily the Centers for Disease Control and Prevention (CDC) and potentially the World Health Organization (WHO), during the COVID-19 pandemic. The goal is to develop a model that could be used to better understand public reactions and opinions, allowing health authorities to more effectively determine which public health measures should be implemented when disseminating information, with the purpose of keeping the population properly informed during public health crises such as pandemics.

This can be accomplished through Natural Language Processing (NLP) techniques, particularly sentiment analysis, which aims to identify emotional patterns, levels of acceptance or rejection, as well as changes in public behavior and perception regarding official announcements and health recommendations. This would be achieved by analyzing comments and user responses on digital platforms and studying how emotions such as fear, trust, frustration, anger, denial, acceptance, pandemic fatigue, and institutional distrust evolved over time.

Additionally, the model could be used to evaluate how different types of official publications such as those related to vaccines, masks, lockdowns, health recommendations, and statistical reports generated different reactions among the population.

2. Problem Statement

Pandemics represent one of the most serious public health risks that health institutions may face. This is largely because, as seen in the case of COVID-19, the virus itself can spread and infect people at a very rapid rate, making it difficult to control while also leaving authorities with limited time to react with appropriate measures. In situations like these, prevention becomes the primary strategy used by public health institutions to combat the disease.

During the COVID-19 pandemic, official public health organizations continuously disseminated information, recommendations, and preventive measures through social media and digital platforms.

"CDC emphasizes core strategies to lower health risks from COVID-19, including severe outcomes such as hospitalization and death. Preventing severe outcomes from COVID-19 illness helps prevent Long COVID. Steps you can take to protect yourself and others from COVID-19 include:

- Staying up to date on COVID-19 vaccination
- Practicing good hygiene (practices like handwashing that improve cleanliness)
- Taking steps for cleaner air
- When you may have a respiratory virus: Taking precautions to prevent spread

Seeking healthcare promptly for testing and/or treatment if you have risk factors for severe illness; treatment may help lower your risk of severe illness." (Center for Disease Control and Prevention, 2026)

However, public reactions to these communications varied significantly, ranging from acceptance and support to distrust, rejection, and social polarization.

In the context of a pandemic, it is essential that epidemiologists and public health authorities encourage the general population to adopt preventive health measures to contain the spread of the disease—one of the most critical measures for bringing the pandemic to an end being vaccination.

"The use of safe and effective vaccines helps ensure that COVID-19 does not cause a severe or fatal form of the disease. It is estimated that, in 2021 alone, COVID-19 vaccines saved the lives of approximately 14.4 million people worldwide. Vaccination also reduces the likelihood of new variants emerging." (World Health Organization, 2023)

Consequently, understanding how the population reacted to these communications is of paramount importance, given that public perception directly influences both compliance with health measures and the acceptance of vaccines. For this reason, this project will use NLP techniques to analyze comments and publications related to COVID-19 in order to identify emotional patterns and changes in collective sentiment over time.

This project is considered highly relevant within the field of NLP because it seeks to support epidemiological prevention efforts by applying sentiment analysis and text classification models to interpret large volumes of unstructured human language data obtained from social media platforms.

3. Dataset Selection

Due to the absence of a pre-existing dataset that fully captures public sentiments in direct response to official COVID-19 publications from health authorities such as the CDC and WHO, this project will require the construction of a custom dataset through targeted data collection and scraping of relevant social media platforms.

To capture the nuances of public sentiment toward official health communications, this project will utilize a multi-source, custom-built dataset. By focusing on direct interactions with official entities, we ensure the data is highly relevant to the problem of institutional trust.

- Data Selection
  - Language: Posts will be exclusively in English language, to ensure consistency in NLP processing
  - Time Range: Aligned with key public health milestones throughout the pandemic including the initial outbreak declaration, mask mandates, vaccine rollout, and lifting of restrictions
  - Relevance: The posts must directly reference or respond to official CDC or WHO publications and announcements.
- Data Sources:
  - Twitter/X: We will scrape approximately 8,000 tweets and replies directly associated with verified accounts (e.g., @CDCgov, @WHO). This will include posts containing specific hashtags such as #COVID19, #VaxUp, and #CDCRecommendations.
- Data Structure: The dataset will be organized into a structured format (CSV/JSON) containing the following attributes:
  - Timestamp: To track emotional evolution across pandemic milestones.
  - Raw_Text: The original content for NLP preprocessing.

The Twitter/X platform is one of the leading platforms, thanks to its "trending topics" feature, which helps identify relevant conversations and commentary at key moments a capability that will be essential for identifying comments during critical stages of the pandemic.

“A word, phrase, or topic that is mentioned at a greater rate than others is said to be a "trending topic". Trending topics become popular either through a concerted effort by users or because of an event that prompts people to talk about a specific topic. These topics help Twitter and their users to understand what is happening in the world and what people's opinions are about it. Websites that track and display trending topics, like TwitterTrend.co, provide real-time information on the most discussed topics worldwide, offering users insights into regional and global trends.” (TwitterTrend.co., 2024)

4. Expected Outcomes and Evaluation Plan

The primary outcome of this project is a robust multi-class classification model capable of detecting shifts in public trust in real-time.

- Anticipated Results: We expect the model to identify "Institutional Distrust" and "Pandemic Fatigue" as the dominant sentiments during late-stage pandemic milestones (e.g., the lifting of mask mandates).
- Model Performance Metrics: To ensure scholarly rigor, the model will be evaluated using:
    - F1-Score: Since emotional categories like "Anger" or "Denial" may be imbalanced in the dataset, the F1-Score will serve as our primary metric to balance precision and recall.
    - Accuracy: Accuracy will provide a general measure of the model's overall classification performance across the dataset.
    - Confusion Matrix: We will utilize a confusion matrix to identify specific "emotional overlaps"—for instance, where the model struggles to distinguish between "Frustration" and "Anger."
- Proposed Deliverables:
    - A Sentiment Trend Map visualizing how public trust fluctuated following major CDC announcements.
    - A Feature Importance Analysis identifying which keywords (e.g., "mandate," "booster," "science") were the strongest predictors of negative sentiment.

Bibliography

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